Forward Return Expectations

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MARCH 2026

Background

Well-studied set of questions:

- What is the expected return on the market from now until $t + 1$?
- How does it evolve over time? Are there systematic errors in return predictions?
- Through the lens of Campbell–Shiller decomp. [low expected returns $\implies$ high prices, all else equal]:

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$$p_t - d_t = \kappa - \underbrace{\sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t r_{t+j+1}} + \sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t \Delta d_{t+j+1}$$

$$\underbrace{\mathbb{E}_t r_{t+1}}_{\text{low-powered tests}} + \underbrace{\textstyle\sum_{j=1}^{\infty} \rho^j \, \mathbb{E}_t r_{t+j+1}}_{\text{much more important for pricing!}}$$

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$$p_t - d_t = \kappa - \underbrace{\sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t r_{t+j+1}}_{\mathbb{E}_t r_{t+1} + \sum_{j=1}^{\infty} \rho^j \, \mathbb{E}_t r_{t+j+1}} + \sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t \Delta d_{t+j+1}$$

Our focus:

- What is the **expected future equity premium**?
- How does it compare to the *actual* future equity premium $\mathbb{E}_{t+j} r_{t+j+1}$?
- Are there systematic errors in *expected* return predictions?

What We Do

1. Using options & surveys, measure return expectations at multiple horizons n (short & long):

$$\textbf{Spot rate (current):} \qquad \mu_t^{(n)} \;=\; \mathbb{E}_t[r_{t,t+n}]$$

2. From these, back out expected future return expectation (*forward return expectation*):

$$\textbf{Forward rate (future):} \qquad f_t^{(n)} \;=\; \mu_t^{(n+1)} - \mu_t^{(n)} \;=\; \mathbb{E}_t[\mu_{t+n}^{(1)}]$$

3. Compare forward rate to realized future spot rate:

$$\textbf{Forecast error:} \qquad \varepsilon_{t+n} \;=\; \mu_{t+n}^{(1)} - f_t^{(n)}$$

- Usually will subtract risk-free rate & consider spot/forward risk premium ($\tilde{\mu}_t^{(n)} = \mathbb{E}_t[r_{t,t+n} - r_{t,t+n}^f]$)
  (paper also has results for expected returns)
- Options: New theoretical results to measure forecast errors in forward rates
- Surveys: Term structure of expected returns in Livingston, Duke CFO, Vanguard

What We Do

$$
\begin{aligned}
\textbf{Spot rate:} \quad & \mu_t^{(n)} &&= \mathbb{E}_t[r_{t,t+n}] \\
\textbf{Forward rate:} \quad & f_t^{(n)} &&= \mu_t^{(n+1)} - \mu_t^{(n)} = \mathbb{E}_t[\mu_{t+n}^{(1)}] \\
\textbf{Forecast error:} \quad & \varepsilon_{t+n} &&= \mu_{t+n}^{(1)} - f_t^{(n)}
\end{aligned}
$$

Measurement:

1. Option prices
   - Theory-based tools to measure forward rates & forecast errors
   - Test whether expectations are intertemporally consistent, without needing to take a stand on whether spot expected returns are themselves rational
   - Rich data...but ultimately model-based
2. Survey expectations
   - Term structure of expected returns in Livingston, Duke CFO, and Vanguard surveys
   - Simple, model-free estimates (will start with these). . .but not as rich data

What We Find

$$
\begin{aligned}
\textbf{Spot rate:} \quad & \mu_t^{(n)} &&= \mathbb{E}_t[r_{t,t+n}] \\
\textbf{Forward rate:} \quad & f_t^{(n)} &&= \mu_t^{(n+1)} - \mu_t^{(n)} = \mathbb{E}_t[\mu_{t+n}^{(1)}] \\
\textbf{Forecast error:} \quad & \varepsilon_{t+n} &&= \mu_{t+n}^{(1)} - f_t^{(n)}
\end{aligned}
$$

Excess countercyclicality in forward return expectations:

1. In options & all surveys, forward rates are consistently countercyclical...
   - When the market $\searrow \implies$ expectations of future equity premia $\nearrow$
   - Contrasts with short-horizon return extrapolation in some surveys
2. ...and in fact **too countercyclical**
   - In bad times, investors believe expected returns will stay elevated for longer and by more than their own subsequent beliefs justify (vice versa in good times)
   - Thus excessively cyclical (and excessively volatile) forward return expectations
   - As if investors **overreact to news about risk**

A Unified View of Subjective Return Expectations

[+ : pos. comovement with objective meas.]	Horizon of expectations				
	Short run	+	Forward *(this paper)*	=	Long run
Sophisticated investors					
"Mr. Market" (options) [Martin]	+		+ (excess)		+
Asset managers [Dahlquist-Ibert]					+
Other professionals					
CFOs [Greenwood-Shleifer, De La O-Myers]	–		+ (excess)		+
Prof. forecasters [Nagel-Xu]	+ (insufficient)		+ (excess)		+
Retail investors					
Vanguard investors [Giglio et al.]	–		+ (excess)		+
Gallup survey [Greenwood-Shleifer]	–				

Implications

Excess cyclicality in forward return expectations helps us understand:

1. Debate on cyclicality of subjective risk premia
   - Short-run expectations differ across investor groups (pro-, a-, or countercyclical)
   - Forward expectations are *consistently* countercyclical — and excessively so
2. Excess volatility in stock prices
   - When prices are depressed, partly reflects investors expecting persistently high risk premia
   - Lower-bound estimates: 20–25% of price declines during 2008 and Covid crises; largest estimates explain the majority of the drops
3. Facts about equity term structure & inelastic demand for equities *(in paper)*
4. Models of belief formation & forecasting in different environments

Explanations: Simple, externally calibrated model of overreaction to risk at long horizons works well

. . .and also explore how it may coexist alongside short-run return extrapolation

Roadmap

1. Introduction

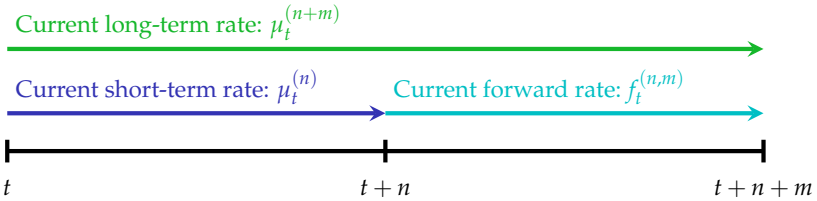
2. Measuring Forward Expectations
   - Identification Challenges
   - Survey-Based Measurement
   - Price-Based Measurement

3. Empirical Tests

4. Forecast Errors: Implications and Explanations

5. Conclusions

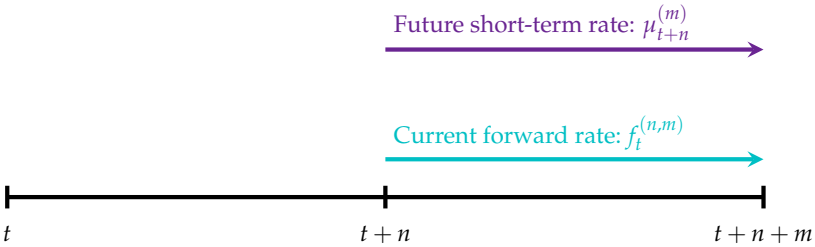
Identification Challenges



Issues:

1. How to measure spot & forward rates for risky assets
   - For risk-free Treasury term structure, rates are observable at multiple horizons

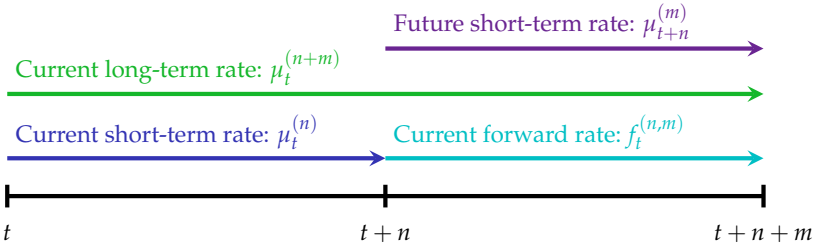
Identification Challenges



Issues:

1. How to measure spot & forward rates for risky assets
   - For risk-free Treasury term structure, rates are observable at multiple horizons
2. For rates measured from market prices, how to deal with risk premium
   - May have $f_t^{(n,m)} \neq \mathbb{E}_t[\mu_{t+n}^{(m)}]$, since price $\neq$ subjective expectation
   - Same issue arises for risk-free term structure (failure of expectations hypothesis)

Identification Challenges



Issues:

1. How to measure spot & forward rates for risky assets
2. How to deal with risk premium

Solutions:

(i) Find surveys that elicit return expectations at multiple horizons

(ii) Use a theoretically motivated set of derivatives prices

Survey-Based Measurement

Three surveys that elicit return expectations at multiple horizons:

1. Livingston survey of professional economists & forecasters, via Philly Fed (1992–2021, $2\times/\text{yr}$)
   - Median S&P 500 price expectations at 6m & 12m horizon, from which:
     - $\mu_t^{(\text{12 months})}, \mu_t^{(\text{6 months})}$
     - $f_t^{(\text{6 months})} = \mu_t^{(\text{12 months})} - \mu_t^{(\text{6 months})}$
     - $\varepsilon_{t+\text{6 months}}^{(\text{6 months})} = \mu_{t+\text{6 months}}^{(\text{6 months})} - f_t^{(\text{6 months})}$
   - When considering risk premia, use T-bill expectations at 6m & 12m horizon as well

Survey-Based Measurement

Three surveys that elicit return expectations at multiple horizons:

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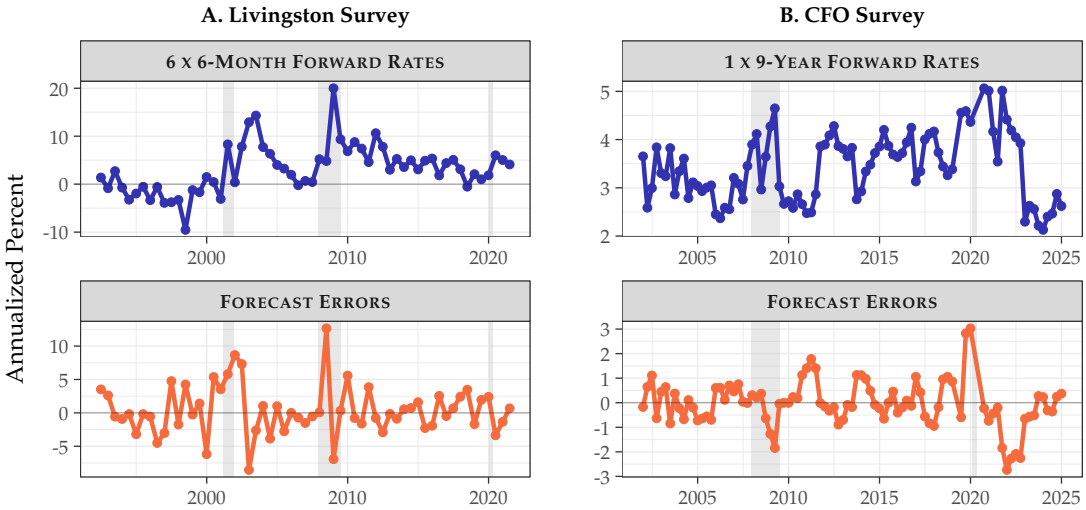
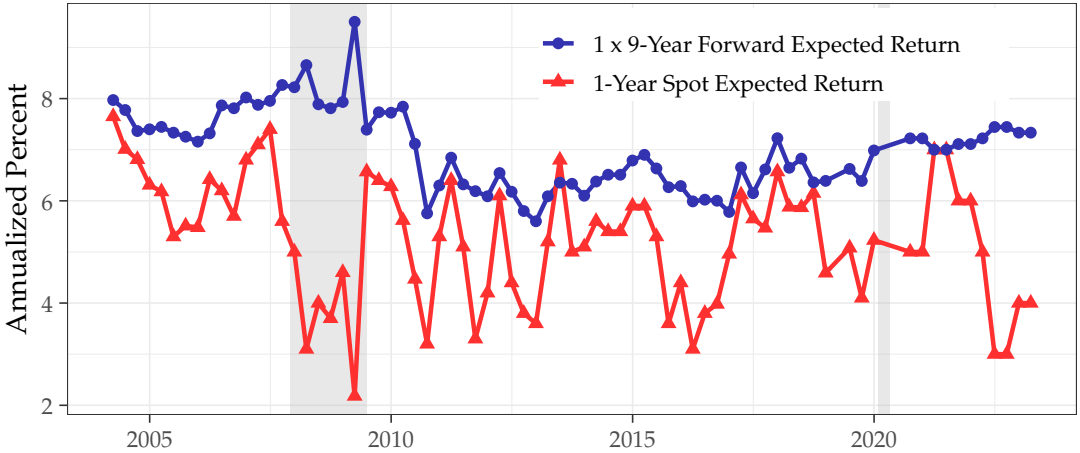
   - Median S&P 500 price expectations at 6m & 12m horizon, from which:
     - $\mu_t^{(\text{12 months})}, \mu_t^{(\text{6 months})}, f_t^{(\text{6m})} = \mu_t^{(\text{12m})} - \mu_t^{(\text{6m})}, \varepsilon_{t+\text{6m}}^{(\text{6m})} = \mu_{t+\text{6m}}^{(\text{6m})} - f_t^{(\text{6m})}$
2. Duke–Fed CFO survey (2001–2025, quarterly; structural survey break in 2020) Break plot
   - Mean 1y & 10y S&P 500 return expectations, from which:
     - $\mu_t^{(\text{10 years})}, \mu_t^{(\text{1 year})}$
     - $f_t^{(\text{1 year, 9 years})} = \mu_t^{(\text{10 years})} - \mu_t^{(\text{1 year})}$
     - $\varepsilon_{t+1\text{ year}}^{(\text{9 years})} \approx \mu_{t+1\text{ year}}^{(\text{10 years})} \times 9/10 - f_t^{(\text{1 year, 9 years})}$
   - For risk premia, subtract observed 10y & 1y yield on Treasuries
3. Vanguard survey of retail investors (2017–2024, bimonthly, ∼2,000 respondents)
   - Mean 1y & 10y expected returns, same construction as CFO survey

Excessively Countercyclical Forward Risk Premia



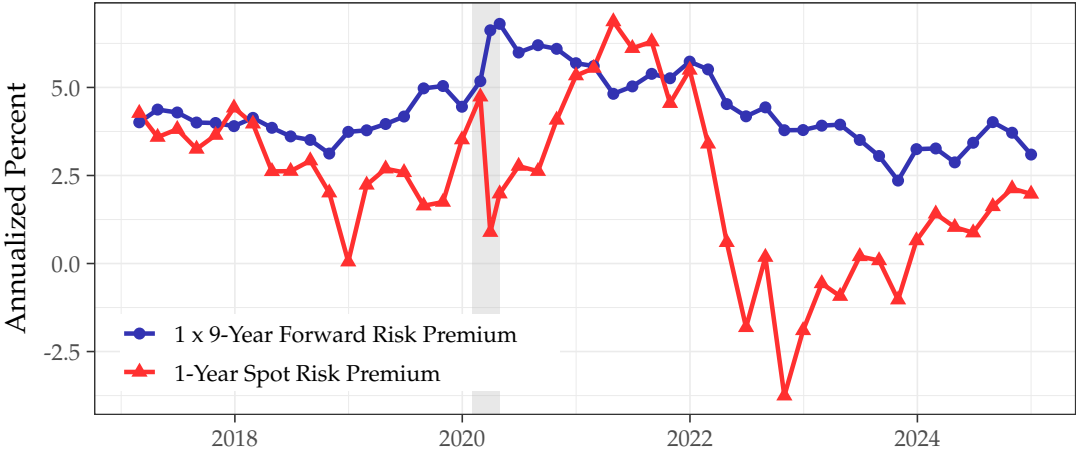
CFO Spot and Forward Rates: Importance of Long Horizon



- CFOs extrapolate past returns at short horizons, but long-horizon forecasts are (too) **countercyclical**
- Countercyclical forwards are stronger for risk premia (previous slide)

Vanguard Retail Survey: Spot and Forward Rates

Same pattern in Vanguard survey of ∼2,000 retail investors (2017–2024):



- Forecast error predictability also holds statistically, as we'll see

Price-Based Measurement for Marginal Investor Expectations

Intuition: w/ option prices, measure $\mathbb{E}[\text{future eq. premium}] + \text{risk prem.}$, and risk prem. is small

- **Building block**: **LVIX** $\mathcal{L}_t^{(n)}$ (Gao & Martin 2021): Given SDF (marginal utility) $M_{t,t+n}$,

$$\underbrace{\mathbb{E}_t[r_{t,t+n}]}_{\substack{\mu_t^{(n)} \\ \textit{[what we want]}}} = \underbrace{\mathbb{E}_t[M_{t,t+n}R_{t,t+n}r_{t,t+n}]}_{\substack{\mathcal{L}_t^{(n)} \\ \textit{[observable RN expectation]}}} - \underbrace{\mathrm{Cov}_t(M_{t,t+n}R_{t,t+n}, r_{t,t+n})}_{\substack{\mathcal{C}_t^{(n)} \\ \textit{[unobservable]}}}$$

- Steps: $\mathbb{E}_t[M_{t,t+n}R_{t,t+n}] = 1 \Rightarrow \mu_t = \mathbb{E}_t[r_{t,t+n}] = \mathbb{E}_t[r_{t,t+n}]\mathbb{E}_t[M_{t,t+n}R_{t,t+n}]$
- $\mathcal{C}_t^{(n)} = 0$ under log utility ($MR = 1$)...otherwise introduces unobservable contamination
- Gao & Martin: $\mathcal{C}_t^{(n)} \leqslant 0$...but what about for fwd rate $f_t^{(n,m)} = \mathcal{L}_t^{(n+m)} - \mathcal{L}_t^{(n)} + \mathcal{C}_t^{(n)} - \mathcal{C}_t^{(n+m)}$?
- **Key insight:** Covariance terms largely cancel when measuring **forecast errors** $\varepsilon_{t+n}^{(m)} = \mu_{t+n}^{(m)} - f_t^{(n,m)}$
- Option-based expected returns may not be good predictors of realized returns...

  ...but they should be **consistent over time**

The Log-Normal Case: Result

Observable forecast-error proxy:

$$\widehat{\varepsilon}_{t+n}^{(m)} = \mathscr{L}_{t+n}^{(m)} - \mathscr{L}_{t}^{(n+m)} + \mathscr{L}_{t}^{(n)}$$

Result 1 (*Log-Normal Identification*)

For a general SDF $M_{t,t+n}$, assuming $M_{t,t+n}$, $R_{t,t+n}$ are jointly log-normal:

$$\mathbb{E}_t\left[\widehat{\varepsilon}_{t+n}^{(m)}\right] = \mathbb{E}_t\left[\varepsilon_{t+n}^{(m)}\right] - \mathrm{Cov}_t(M_{t,t+n}R_{t,t+n}, \mathbb{E}_{t+n}[r_{t+n,t+n+m}])$$

- Covariance term now relates to pricing of *discount-rate risk*, rather than *realized-return* risk
- Likely much smaller than previous term: expected returns are much less volatile than realized returns
- Can be disciplined empirically or theoretically
- As long as covariance is small, can use LVIX to measure fwd-rate forecast errors & test $\mathbb{E}_t[\varepsilon_{t+n}^{(m)}] = 0$
- Basic idea of proof: $MR_{t,t+n}$ is orthogonal to unexpected component of $r_{t+n,t+n+m}$

The General Case: Result

Define forecast-error proxy and expected-return proxy:

$$\widehat{\varepsilon}_{t+n}^{(m)} = \mathscr{L}_{t+n}^{(m)} - \mathscr{L}_{t}^{(n+m)} + \mathscr{L}_{t}^{(n)}$$

$$\widehat{\mu}_{t+n}^{(m)} = \mathscr{L}_{t+n}^{(m)} + r_{t+n,t+n+m}^{f}$$

Result 2 (*Generalized Identification*)

For any SDF $M_{t,t+n}$,

$$\mathbb{E}_t\left[\widehat{\varepsilon}_{t+n}^{(m)}\right] = \mathbb{E}_t\left[\varepsilon_{t+n}^{(m)}\right] - \mathrm{Cov}_t\left(M_{t,t+n}R_{t,t+n}, \widehat{\mu}_{t+n}^{(m)}\right)$$

- Intuition from log-normal case carries over, with $\mathbb{E}_{t+n}[r_{t+n,t+n+m}]$ replaced by $\widehat{\mu}_{t+n}^{(m)}$
- LVIX-based $\widehat{\mu}_{t+n}^{(m)}$ is closely related to $\mathbb{E}_{t+n}[r_{t+n,t+n+m}]$ . . .but $\widehat{\mu}_{t+n}^{(m)}$ is directly observable
- Main specification: $\widehat{\mu}_{t+n}^{(m)}$ is $\frac{1}{10}$ as volatile as realized return $r_{t+n,t+n+m}$

  $\Longrightarrow$ unobserved covariance likely smaller for forecast errors than for spot rates

Roadmap

1. Introduction

2. Measuring Forward Expectations

3. Empirical Tests

4. Forecast Errors: Implications and Explanations

5. Conclusions

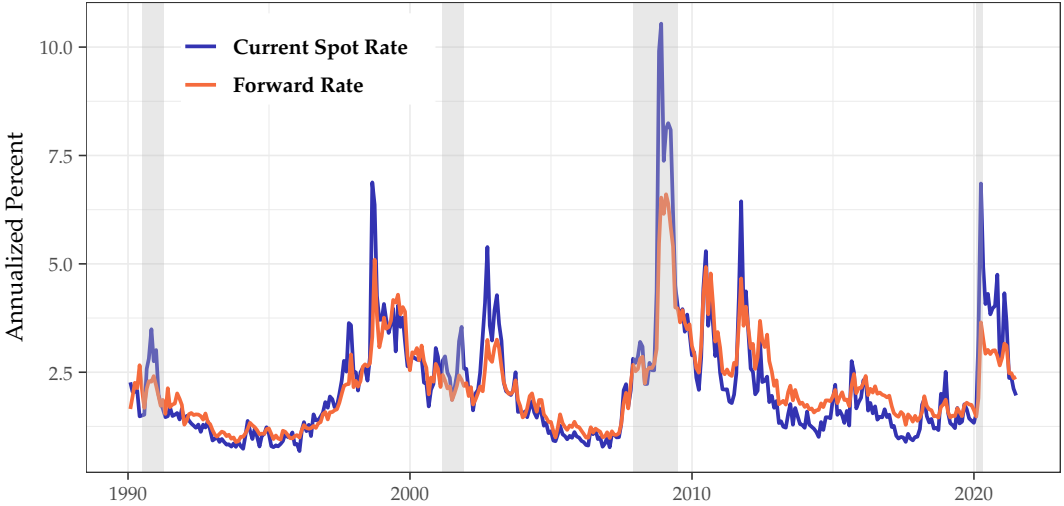
Data and Measurement: Options

Data:

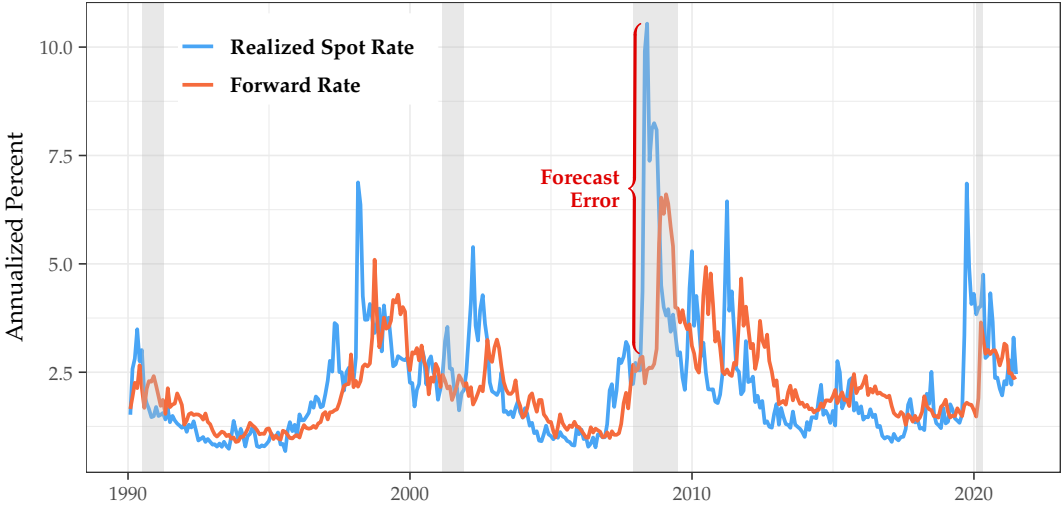
- Main data: Global panel of index options from OptionMetrics (monthly data, standard filters)
  - For U.S. sample: 1990–2021
  - For international sample: Consider 10 major indices, with data since at least 2006
- Baseline: 6-month horizon, 6 months forward ($n = m = 6$)
- Measuring LVIX: Integrate price$(K)/K$ following Carr & Madan (2001)
  [extrapolate past observed strikes using a bunch of different methods]

- Calculate integral a bunch of different ways
- First: Simplify by working under log assumption, so LVIX $\Longrightarrow$ spot & forward rates

Estimates: Contemporaneous U.S. Spot and Forward Rates



Estimates: Realized U.S. Spot and Forward Rates



Do Forward Rates Predict Future Spot Rates?

Mincer–Zarnowitz Regressions for Future Spot Rates

$$\tilde{\mu}_{t+6}^{(6)} = \beta_0 + \beta_1 \tilde{f}_t^{(6,6)} + \epsilon_{t+6}$$

	(1) U.S.	**(2)** Ex-U.S.	**(3)** All
$f_t^{(6,6)}$	0.67***	0.55***	0.56***
	(0.096)	(0.056)	(0.055)
Intercept	0.74***		
	(0.28)		
Country FEs	✗	✓	✓
p-value: $\beta_1 = 1$	0.003	0.000	0.000
Obs.	378	1,849	2,227
R^2	0.22	0.21	0.22
Within R^2	—	0.14	0.15

SEs in (1) are heteroskedasticity and autocorrelation-robust [Lazarus et al. (2018)], and in (2)–(3) are clustered by exchange and month.

- Substantial predictive power...
  ...but $\beta_1 \neq 1$, so forward rates overshoot future spot rates
- The market **qualitatively** understands variation in the equity premium, but **quantitatively** significant excess forward volatility
- What if β_1 estimate is downwardly biased due to measurement error?
- Consider IV using shorter-term forward rate $f_t^{(2,1)}$ (shorter horizons likely better measured)

Do Forward Rates Predict Future Spot Rates?

Instrumented Mincer–Zarnowitz Regressions for Spot Rates

$$\tilde{\mu}_{t+6}^{(6)} = \beta_0 + \beta_1 \tilde{f}_t^{(6,6)} + \epsilon_{t+6}, \quad \tilde{f}_t^{(6,6)} = \pi_0 + \pi_1 \tilde{f}_t^{(2,1)} + \eta_t$$

	(1) U.S.	(2) Ex-U.S.	(3) All
$f_t^{(6,6)}$	0.73***	0.69***	0.70***
	(0.062)	(0.078)	(0.074)
Intercept	0.59***		
	(0.13)		
Country FEs	✗	✓	✓
p-value: $\beta_1 = 1$	0.018	0.004	0.003
Obs.	378	1,849	2,227
R^2	0.22	0.20	0.22
Within R^2	—	0.13	0.14

SEs in (1) are heteroskedasticity and autocorrelation-robust [Lazarus et al. (2018)], and in (2)–(3) are clustered by exchange and month.

- Forward rate ↗ by **1%** $\Longrightarrow$ future spot rate ↗ by ∼**0.7%**
- Forward rates explain ∼**20%** of the variation in future spot rates
- Again: The market **qualitatively** understands variation in the equity premium, but **quantitatively** significant excess forward volatility

Average Forecast Errors Are Close to Zero

Average Forecast Errors Across Countries

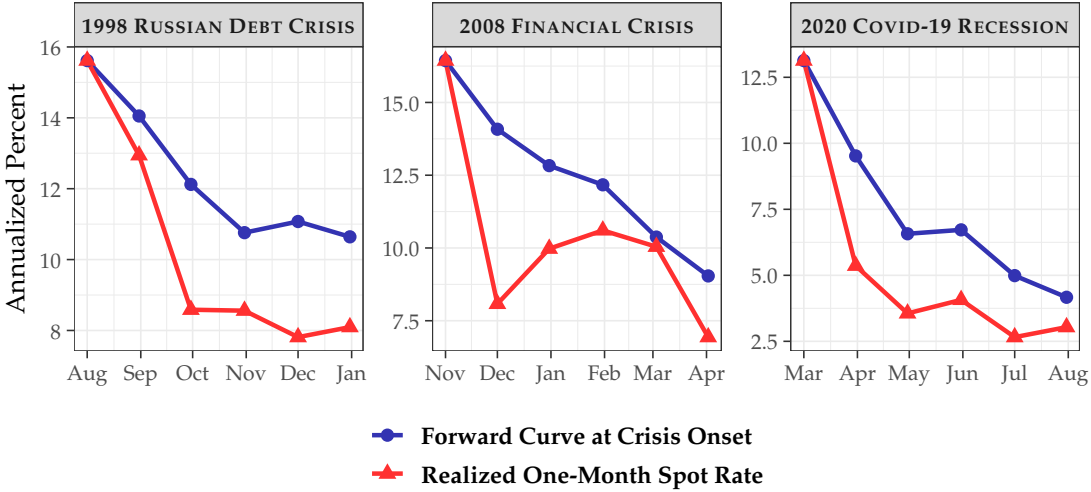
$$\tilde{\varepsilon}_{t+6}^{(6)} = \tilde{\mu}_{t+6}^{(6)} - \tilde{f}_t^{(6,6)}$$

	(1) U.S.	**(2)** Ex-U.S.	**(3)** All
Average	0.021	0.20	0.17
	(0.15)	(0.11)	(0.11)
Obs.	378	1,849	2,227

SEs in (1) are HAR [Lazarus et al. (2018)], and in (2)–(3) are clustered by exchange and month.

- Not just statistically insignificant, but effectively zero: $\bar{\varepsilon} \leqslant 20$ bps annualized
- But average of zero masks substantial predictability...

Option-Based Forward and Realized Spot Rates in Crises



. . .and pattern holds systematically in both options & surveys.

Forward Rates as Predictors of Forecast Errors

Predictable Forecast Errors

Regressions of Forecast Errors on 2×1 Forward Rate

$$\tilde{\varepsilon}^{(6)}_{t+6} = \beta_0 + \beta_1 \tilde{f}^{(2,1)}_t + e_{t+6}$$

	(1) U.S.	(2) Ex-U.S.	(3) All
$f_t^{(2,1)}$	-0.17**	-0.16**	-0.16***
	(0.066)	(0.049)	(0.047)
Intercept	0.39*		
	(0.23)		
Country FEs	✗	✓	✓
Obs.	378	1,849	2,227
R^2	0.04	0.04	0.04
Within R^2	—	0.03	0.03

SEs in (1) are HAR [Lazarus et al. (2018)], and in (2)–(3) are clustered by exchange and month.

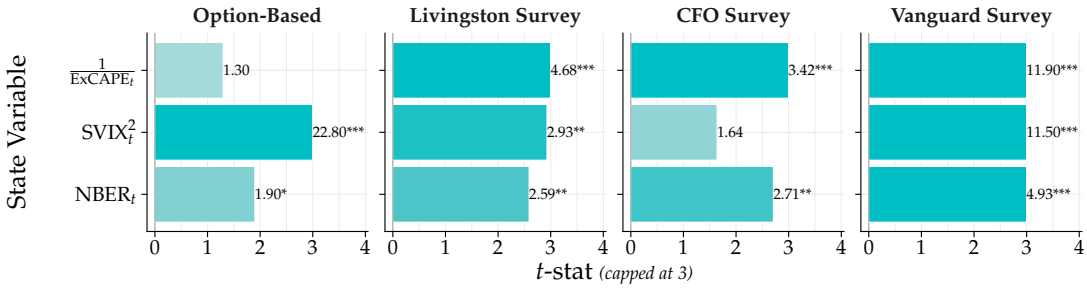
- Forward rates again overshoot future spot rates
- Errors are also predictable in Coibion–Gorodnichenko regressions using forward-rate *revisions*
- Is this consistent with overreaction? It depends: Overreaction to what?
  - News about risk/risk premia: Yes
  - Past returns: Wrong direction
  - Consistent excess cyclicality

Consistent Evidence From Options and Surveys

	Expectations Measured by:			
	Options	Livingston Survey	CFO Survey	Vanguard Survey
Panel A: Predictability of Spot Rates $(\tilde{\mu}_{t+1} = \beta_0 + \beta_1 \tilde{f}_t + e_{t+1})$				
β_1	0.67***	0.81*	0.35***	0.43**
R^2	0.22	0.56	0.59	0.13
Panel B: Predictability of Forecast Errors $(\tilde{\varepsilon}_{t+1} = \beta_0 + \beta_1 \tilde{f}_t + e_{t+1})$				
β_1	-0.17**	-0.19*	-0.65***	-0.57**
R^2	0.04	0.05	0.48	0.21

U.S. data. Options/Livingston: $6 \times 6m$. CFO/Vanguard: $1y \times 9y$. SEs are HAR [Lazarus et al. (2018)], and significance stars in Panel A are for test of $\beta_1 = 1$.

Robustly Countercyclical Forward Risk Premia



- All forward risk premia comove with standard countercyclical indicators: inverse valuation ratio (1/ExCAPE), option-implied volatility (SVIX), and NBER recession indicator
- So for all investor groups, expected future equity premia go up concurrently in bad times, and by too much relative to future views on actual equity premium

Roadmap

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3. Empirical Tests

4. Forecast Errors: Implications and Explanations
   - Implications & Excess Volatility in Crises
   - Explanations & A Simple Model

5. Conclusions

Implications

Excess cyclicality in forward return expectations helps us understand:

1. Debate on cyclicality of subjective risk premia
   - Short-term return expectations sometimes appear procyclical [Greenwood & Shleifer 2014], acyclical [Nagel & Xu 2022], or countercyclical [Dahlquist & Ibert 2024]
   - But long-horizon forward expectations are countercyclical across all data sources (excessively so)
   - Disagreement stems in part from differences in horizon
   - Our focus on forward rates eliminates disagreement across data sources

Implications

Excess cyclicality in forward return expectations helps us understand:

2. Excess volatility in stock prices
   - When prices are depressed, this partly reflects investors expecting persistently high risk premia
   - If investors didn't overestimate persistence, would see more modest fluctuations in prices
   - Next slides: Quantification

How Significant Are Forecast Errors for Price Variation?

$$p_t - d_t = \kappa - \sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t r_{t+j+1} + \sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t \Delta d_{t+j+1}$$

How Significant Are Forecast Errors for Price Variation?

$$p_t - d_t = \kappa - \mathbb{E}_t r_{t+1} - \sum_{j=1}^{\infty} \rho^j f_t^{(j,1)} - \underbrace{RF_t}_{\substack{\text{discounted} \\ \text{risk-free rates}}} + \sum_{j=0}^{\infty} \rho^j \, \mathbb{E}_t \Delta d_{t+j+1}$$

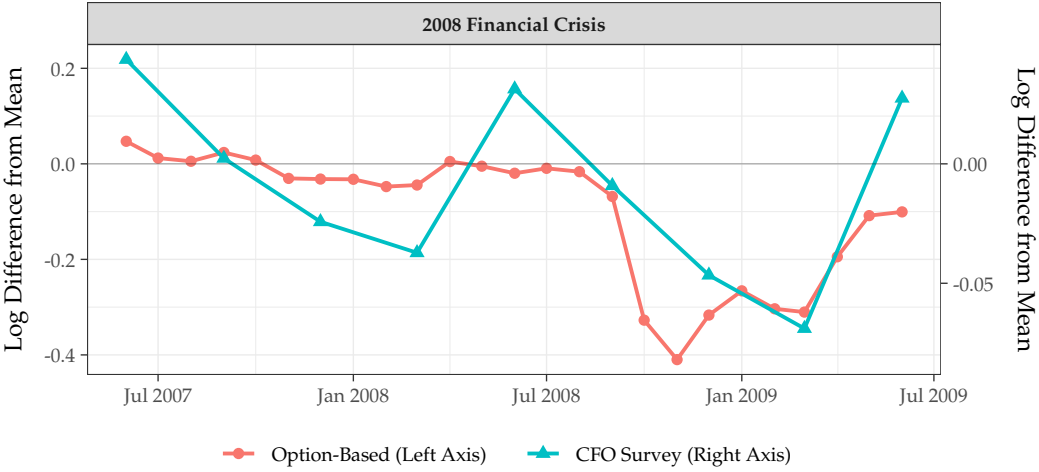
Break $f_t^{(j,1)}$ into:

$$f_t^{(j,1)} = \underbrace{\mathbb{E}_t[\mu_{t+j}^{(1)}]}_{\text{expected spot rates}} - \underbrace{\mathbb{E}_t[\varepsilon_{t+j}^{(1)}]}_{\text{predictable forecast errors}}$$

Call $\mathcal{E}_t \equiv \sum_{j=1}^{\infty} \rho^j \mathbb{E}_t[\tilde{\varepsilon}_{t+j}^{(1)}]$ discounted forecast errors, which have 1:1 impact on prices. Two measures:

1. **Options:** Use baseline predicted forecast errors
   - Estimate decay parameter in $\mathbb{E}_t[\varepsilon_{t+j+1}^{(1)}] = \phi^j \mathbb{E}_t[\varepsilon_{t+j}^{(1)}] \Longrightarrow \widehat{\phi} \approx 1$ *(using longer-dated SX5E data)*
   - Using this, $\widehat{\mathcal{E}}_t^{\text{option}} = \frac{\rho}{1-\rho} \mathbb{E}_t[\varepsilon_{t+1}^{(1)}]$
2. **Surveys:** CFO (2008), Vanguard (2020)
   - No scaling required: estimate first 9 years of forecast errors from predictable component of $\varepsilon_{t+1}^{(9)}$
   - $\widehat{\mathcal{E}}_t^{\text{survey}} = \sum_{j=1}^{9} \rho^j \mathbb{E}_t[\varepsilon_{t+j}^{(1)}] \leq \mathcal{E}_t$, with $\mathbb{E}_t[\varepsilon_{t+j}^{(1)}] = \mathbb{E}_t[\varepsilon_{t+1}^{(9)}]$ annualized

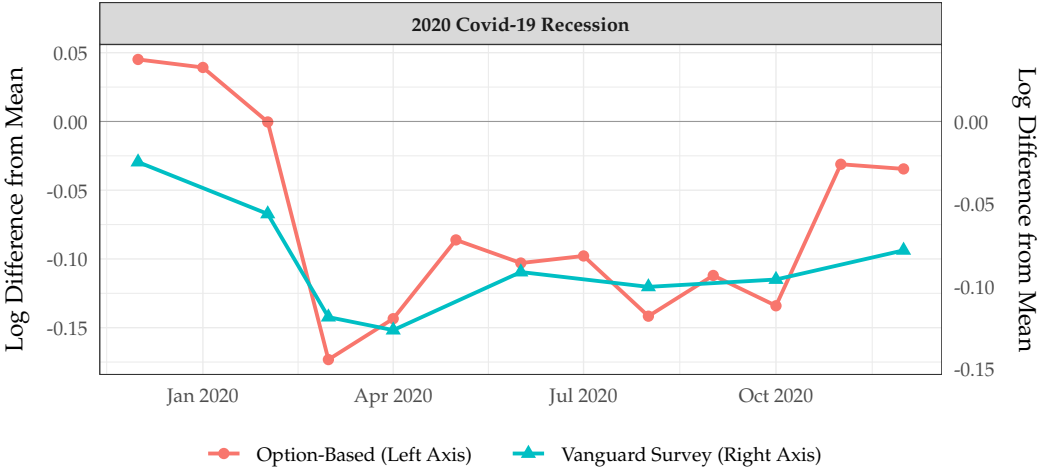
Price Impact of Forecast Errors in Crises



Overestimated future risk premia account for 11% (survey) to 35% (options) of 43% '08 price drop

Less strong outside crises, but big impact in bad times $\Rightarrow$ contributor to excess vol.

Price Impact of Forecast Errors in Crises



Overestimated future risk premia account for 9% (survey) to 22% (options) of 21% '20 price drop

Less strong outside crises, but big impact in bad times $\Rightarrow$ contributor to excess vol.

Further Implications

Excess cyclicality in forward return expectations helps us understand:

3. Facts about equity term structure from dividend claims
   - Risk premia higher than expected $\Longrightarrow$ neg. realized returns, especially for long-duration assets
   - Can generate downward sloping equity term structure on average [Binsbergen, Brandt, Koijen 2012] & upward sloping term structure during bad times (countercyclical variation) [Gormsen 2021]
   - Illustrative quantification Plot
4. More speculatively: Inelastic demand for equities [Gabaix & Koijen 2022]
   - Two possibilities given price drop: Large change in return expectations at short horizons, or moderate change in return expectations at all horizons (our finding)

   $\Longrightarrow$ shouldn't see big increase in current portfolio weight, partially resolving puzzle

Explaining Forecast Errors

Paper considers a few alternative explanations for forecast errors:

1. RE + risk premium Details
   - Price of discount-rate risk must be highly volatile and countercyclical for this to work
   - E.g., if $\text{Corr}_t(r_{t+1}, \mathbb{E}_{t+1} r_{t+2}) = -1$ and negative correlation condition (Gao & Martin 2021) $\Longrightarrow$ **relevant SDF-related covariance can't change sign**
   - Doesn't work for survey evidence
2. Aggressive policy responses that were **rationally unexpected ex ante** Details
   - Ex post monetary policy path surprises are significantly related to ex post forecast errors...
   - ...but explain $\leqslant 10\%$ of their variation
3. **Expectation errors**
   - Simple model: Extrapolation of current level of risk for long-term mean $\Rightarrow$ fwd rates overreact
   - Short-rate bias: Incorrect map from risk to current short rate (allowing short-hor. ret. extrap.)
   - Matches both excess cyclicality in fwd rates & diff. spot-rate cyclicality across investors

A Simple Model of Expectation Errors

- Builds on Hirshleifer, Lim, Teoh (2015) and similar to diagnostic model (BGLS 2019) for fwd rates
- Countercyclical state variable x_t — think of as market variance — follows AR(3)
- Short rate: $\mathbb{E}_t[\mu_t^{(3)}] = \alpha_0 + \alpha_1 x_t$ with $\alpha_1 > 0$ (countercyclical)
- **Bias 1 — Forward rate bias θ_F:** Investors overreact to news about long-run mean $\bar{x}$:

$$\mathbb{E}_t^{\theta}[\bar{x}] = \mathbb{E}_{t-1}^{\theta}[\bar{x}] + \theta_F \left(x_t - \mathbb{E}_{t-1}^{\theta}[\bar{x}] \right)$$

- **Bias 2 — Short rate bias θ_S:** Investors misperceive current short rate:

$$\mathbb{E}_t^{\theta}\left[\mu_t^{(3)}\right] = \mathbb{E}_t\left[\mu_t^{(3)}\right] + \alpha_1 \theta_S \, e_t$$

- θ_F governs excess cyclicality in forwards (**same for all investors**)
- θ_S governs cyclicality of spot rates (**differs across investor groups**)

Model vs. Data and Broader Lessons

- Calibrate x_t to option-implied variance in U.S. sample, and calibrate mapping to spot rates
- $\theta_F = 0.0505$ matches both MZ and error-predictability slopes for options *(similar to past estimates)*
  Plots
- Model also explains heterogeneous spot-rate cyclicality:
  - $\theta_S < 0$: procyclical spot rates (e.g., CFOs, retail — extrapolation)
  - $\theta_S \approx 0$: acyclical spot rates (e.g., professional forecasters)
  - $\theta_S > 0$: countercyclical spot rates (e.g., "Mr. Market")
- **Connection to broader literature on belief formation:**
  - Our results: Overreaction to news about risk premia (low persistence, little long-run news)
  - Risk-free rates have high persistence & **underreaction** there [Farmer, Nakamura, Steinsson 2024]
  - One unification: Overinference from weak signals, underinference from strong [Augenblick, Lazarus, Thaler 2025]

Implied Persistence in Equity-Premium and T-Bill Forecasts

Regressions of Forecasted and Realized Values on Current Values

$$\mathbb{E}_t[\mu_{t+n}] = \alpha_{\text{subj}} + \hat{\rho}\,\mu_t + e_{t,\text{subj}}, \quad \mu_{t+n} = \alpha_{\text{obj}} + \rho\,\mu_t + e_{t,\text{obj}}$$

	Eq. Prem. Forecasts		T-Bill Forecasts
	Options	Livingston Survey	SPF Survey
$\hat{\rho}$	0.65	0.68	0.84
	[0.02]	[0.08]	[0.04]
ρ	0.48	0.64	0.91
	[0.05]	[0.10]	[0.03]

- Overreaction for equity-premium forecasts, underreaction for yield forecasts
- Appears to be **partial** shrinkage to moderate persistence. Consistent with forecasters understanding directional impact of news, but not knowing precise strength.

Roadmap

1. Introduction

2. Measuring Forward Expectations

3. Empirical Tests

4. Forecast Errors: Implications and Explanations

5. Conclusions

Final Notes

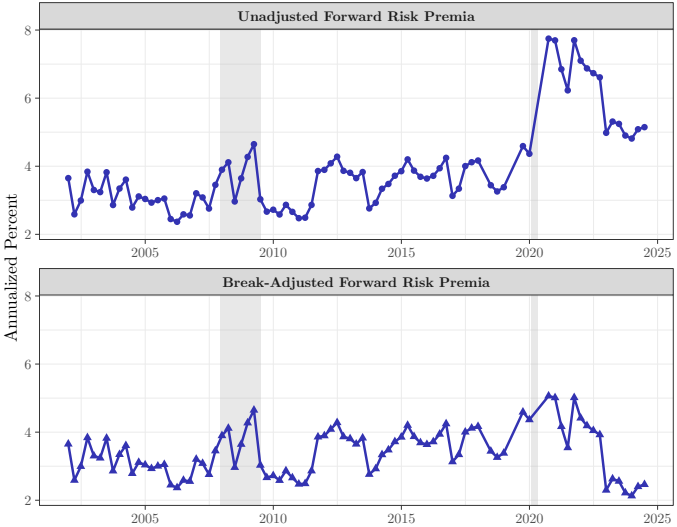
Summary:

- Introduce new methods to measure term structure of expected equity premia
- Robust evidence of excess countercyclicality in forward return expectations
- Investors consistently overestimate how long their own expected returns will stay elevated during bad times, and vice versa during good times
- Consistent across investor groups: options and three surveys (Livingston, CFO, Vanguard)

Thank you!

Appendix

CFO Forward Rates: 2020 Break from Survey Admin. Change



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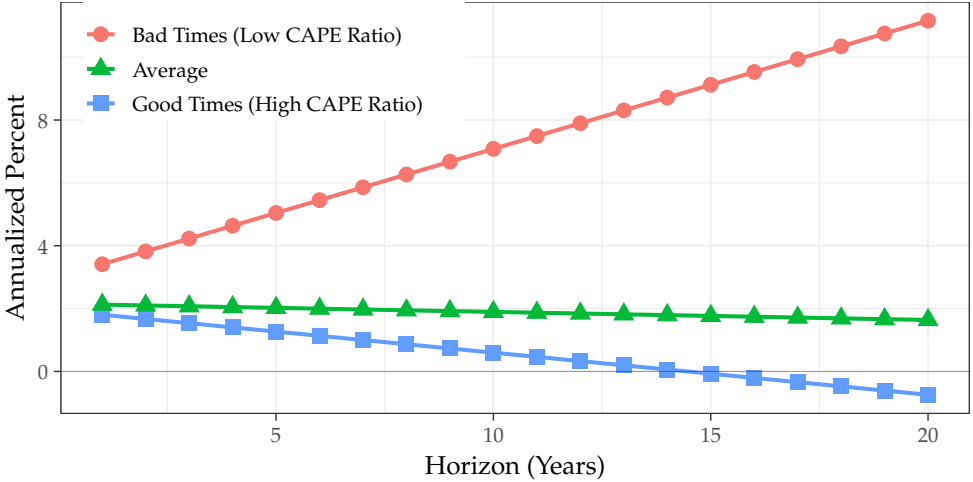
Forecast Errors: Estimated Effect on Returns for Equity Term Structure



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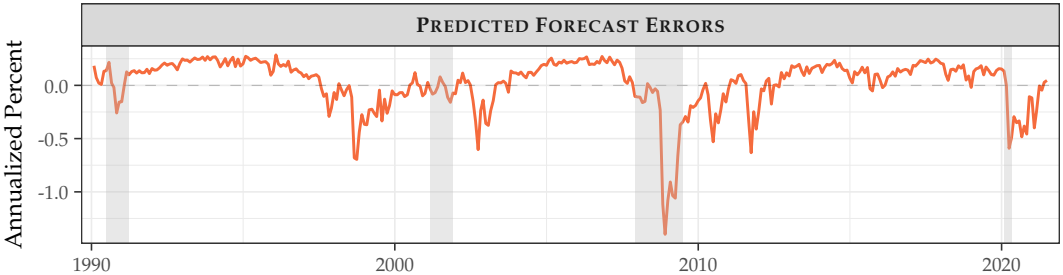
Can Forecast Errors From Price-Based Measure Be Rationalized?

$$\mathbb{E}_t[\widehat{\varepsilon}^{(m)}_{t+n}] = \mathbb{E}_t[\varepsilon^{(m)}_{t+n}] - \underbrace{\mathrm{Cov}_t(MR_{t,t+n}, \mathbb{E}_{t+n}[r_{t+n,t+n+m}])}_{\varsigma_t}$$

What conditions do we need on ς_t in order for **expectation errors** $\mathbb{E}_t[\varepsilon^{(m)}_{t+n}]$ to be unpredictable?
Must have $-\varsigma_t$ take same sign as pred. forecast errors:



Main challenge: Small on average, but must flip signs dramatically ($-$ in good times, $+$ in bad).

Can Forecast Errors From Price-Based Measure Be Rationalized?

$$\mathbb{E}_t[\widehat{\varepsilon}^{(m)}_{t+n}] = \mathbb{E}_t[\varepsilon^{(m)}_{t+n}] - \underbrace{\mathrm{Cov}_t(MR_{t,t+n}, \mathbb{E}_{t+n}[r_{t+n,t+n+m}])}_{\varsigma_t}$$

What conditions do we need on ς_t in order for **expectation errors** $\mathbb{E}_t[\varepsilon^{(m)}_{t+n}]$ to be unpredictable?

- For simplicity: Take $n = m = 1$, and assume SDF and returns are jointly log-normal
- Then $\varsigma_t > 0$ (as needed in bad times) if and only if:

$$SR_t(-\mu_{t+1}) > -\rho_t(r, \mu)\sigma_t(r),$$

  where SR_t is Sharpe ratio on claim to next period's negative equity premium (low payoff is bad)
- Correlation $\rho_t(r, \mu)$ likely to be negative; for illustration, set it to -1
- Then SR_t must vary *more than* $\sigma_t(r)$ for ς_t to flip signs
- One calibration: Go back to log utility (likely to be conservative for time variation in σ_t), and estimate $\sigma_t(r)$ from options

Can Forecast Errors From Price-Based Measure Be Rationalized?

$$\mathbb{E}_t[\widehat{\varepsilon}^{(m)}_{t+n}] = \mathbb{E}_t[\varepsilon^{(m)}_{t+n}] - \underbrace{\mathrm{Cov}_t(MR_{t,t+n}, \mathbb{E}_{t+n}[r_{t+n,t+n+m}])}_{\varsigma_t}$$

What conditions do we need on ς_t in order for **expectation errors** $\mathbb{E}_t[\varepsilon^{(m)}_{t+n}]$ to be unpredictable?

- SR_t must vary *more than* $\sigma_t(r)$ for ς_t to flip signs
- One calibration: Go back to conservative log utility case, and estimate $\sigma_t(r)$ from options. **Results for conditional volatility of 6-month market return:**

Can Forecast Errors From Price-Based Measure Be Rationalized?

$$\mathbb{E}_t[\widehat{\varepsilon}_{t+n}^{(m)}] = \mathbb{E}_t[\varepsilon_{t+n}^{(m)}] - \underbrace{\mathrm{Cov}_t(MR_{t,t+n}, \mathbb{E}_{t+n}[r_{t+n,t+n+m}])}_{\varsigma_t}$$

What conditions do we need on ς_t in order for **expectation errors** $\mathbb{E}_t[\varepsilon_{t+n}^{(m)}]$ to be unpredictable?

- SR_t must vary *more than* $\sigma_t(r)$ for ς_t to flip signs
- Further, given $\rho_t(r, \mu) = -1$, ς_t **cannot** flip signs if mNCC [Gao & Martin (2021), Assumption 2] holds
  - $\rho_t(r, \mu) = -1 \implies \varsigma_t$ is scaled version of their covariance term $\mathcal{C}_t^{(n)}$
  - If $\mathcal{C}_t^{(n)} \leqslant 0$ (mNCC), then $\varsigma_t \leqslant 0$
- More generally, difficult to get both average errors (small) *and* time variation (large) right
- Paper has one illustration varying risk aversion γ

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Unexpected Policy Shocks: Significant, But Limited

Regressions of Forecast Errors on Ex Post Policy Shocks

$$\varepsilon_{t+6}^{(6)} = \beta_0 + \beta_1 \left(\sum_{h=1}^{6} \mathrm{MPS}_{t+h} \right) + e_{t+6}$$

	Explained Variation (R^2)	
	Risk-Free	Risk Premium
BERNANKE-KUTTNER (2005)		
Fed Funds Rate Change	0.40***	0.09**
Fed Funds Rate Surprise	0.01	0.07***
GURKAYNAK-SACK-SWANSON (2005)		
Target Shock	0.05***	0.03*
Path Shock	0.26	0.06**
NAKAMURA-STEINSSON (2018)		
Fed Funds Rate Shock	0.03	0.03**
Policy News Shock	0.32***	0.10**

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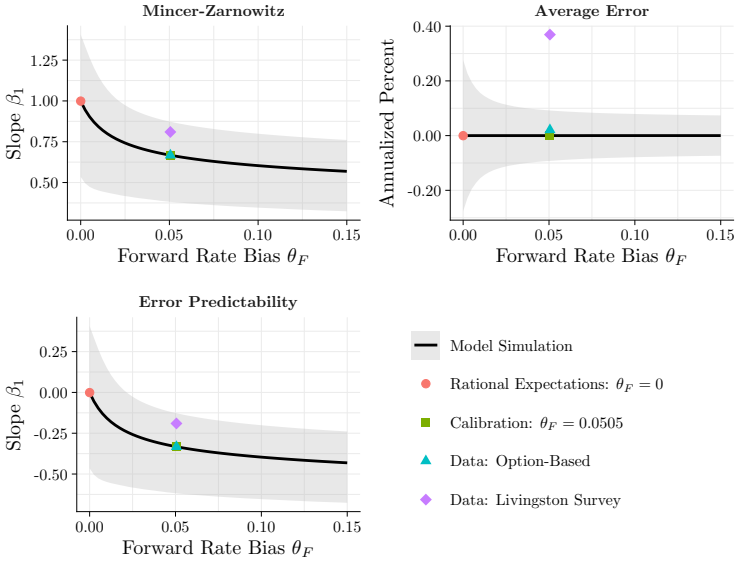
Model vs. Data: Estimates



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A Trilemma for Expectation Errors

$$p_t - d_t = \kappa - \mathbb{E}_t r_{t+1} - \underbrace{\sum_{j=1}^{\infty} \rho^j f_t^{(j,1)}}_{\mathcal{F}_t} + \underbrace{\sum_{j=0}^{\infty} \rho^j \mathbb{E}_t \Delta d_{t+j+1}}_{CF_t} - RF_t$$

- Use $\widetilde{\cdot}$ to denote **expectation error wedge** (deviation from RE economy):

$$\mathrm{var}\left(\widetilde{p_t - d_t}\right) = \mathrm{var}\left(\widetilde{\mathcal{F}}_t\right) + \mathrm{var}\left(\widetilde{CF}_t\right) - 2\,\mathrm{cov}\left(\widetilde{\mathcal{F}}_t, \widetilde{CF}_t\right)$$

- Have to choose between **two of three**:
  1. Volatile expectation errors for cash flows and/or returns
  2. Volatile price-dividend ratio relative to RE
  3. Positive comovement between fundamental and return expectation errors